TRAFFIC SIGNAL DESIGN

CE-529

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1. Introduction

The operational analysis of signalized intersections plays a critical role in modern traffic engineering, especially in urban and suburban environments characterized by rapidly increasing traffic demand and complex driver behavior. The intersection of Burlington Mills Road and New Falls of Neuse Road in North Carolina represents a typical suburban arterial intersection experiencing operational challenges during peak hours due to heavy vehicular volumes, varied turning movements, and the lack of an optimized signal timing plan.

This project is conducted as part of the requirements for CE 529 – Advanced Traffic Engineering at California State University, Long Beach. The course emphasizes practical traffic operational studies, queuing analysis, signal optimization, and the application of traffic simulation and analysis software. The key objective of this project is to evaluate the existing operational conditions at the Burlington Mills and New Falls intersection during the AM peak hour, utilizing data-driven methodologies aligned with the Highway Capacity Manual (HCM 7th Edition) framework.

The study is intended to apply concepts from isolated intersection analysis and signal timing design principles learned throughout the course. A pre-timed traffic signal will be designed for the intersection based on the analysis results. Pre-timed signals are commonly deployed in areas where traffic patterns remain relatively stable throughout the day or where detection technologies are not feasible or cost-effective. The signal design aims to enhance operational efficiency, reduce control delays, and provide acceptable levels of service (LOS) to all user groups.

The analysis focuses on assessing the critical operational measures of effectiveness (MOEs), including control delay, volume-to-capacity ratio, queue lengths, and level of service (LOS) for individual lane groups and approaches. The data inputs, including turning movement counts, peak hour factors (PHFs), saturation flow rates, and geometric configurations, were used from the materials or assumed based on standard engineering practice and HCM recommendations.

Additionally, this study reflects the broader learning objectives of CE 529 — particularly the development of professional skills in traffic signal control design, application of computational tools, and communication of technical findings. The outcome of this project will be a set of optimized pre-timed signal timings, supported by rigorous analysis, which can serve as a preliminary design framework for real-world implementation or further analysis using traffic simulation software.

2. Methodology

2.1 Burlington Mills

The first analysis was performed for the Burlington Mills Road intersection in North Carolina for the AM, Lunch, and PM peak hours to evaluate the existing operational conditions and to design a pre-timed traffic signal following the procedures outlined in the Highway Capacity Manual (HCM 7th Edition). Traffic volume data was provided separately for all three peak periods and was analyzed individually to determine the operational performance of the intersection.

The flow rates for each movement were calculated by applying the PHF to the observed turning movement counts for each peak hour. Saturation flow rates were determined using the HCM base value of 1900 passenger cars per hour per lane (pc/h/ln) with applicable adjustment factors based on geometric features and site-specific conditions. Volume-to-capacity (v/c) ratios and control delays were calculated for each lane group to evaluate the LOS. A pre-timed signal was designed for each peak period using critical movement analysis, phase assignment, and optimized cycle length determination based on desired operational performance.

The following assumptions were made consistently for the AM, Lunch, and PM peak hour analyses:

- *Base Saturation Flow Rate (So)*: 1900 pc/h/ln (HCM standard).
- *Lane Width Factor (f_w)*: 1.04 (lanes wider than 12 feet).
- *Heavy Vehicle Factor (f_{HVg})*: 0.964 – 0.976 (varies slightly by approach and peak period).
- *Left Turn Factor (f_{LT})*: 0.952 (exclusive left-turn lanes).
- *Right Turn Factor (f_{RT})*: 0.847 (exclusive right-turn lanes).
- *Parking Factor (f_p)*: 1.0 (no on-street parking).
- *Bus Blockage Factor (f_{bb})*: 1.0 (no bus stops within the influence area).
- *Lane Utilization Factor (f_{LU})*: 1.0 (balanced lane usage).
- *Work Zone Factor (f_{wz})*: 1.0 (no temporary work zones).

The existing intersection geometry and lane configurations were consistent across all peak periods:

- Southbound Through (SBT) movement has 5 lanes.
- Northbound Through (NBT) movement has 4 lanes.

- Eastbound Left (EBL) and Westbound Left (WBL) movements have 2 dedicated lanes each.
- Protected left-turn phasing was provided for NBL and SBL due to 55 mph posted speed limits.
- Protected left-turn phasing was provided for EBL and WBL due to the presence of dual left-turn lanes.
- No pedestrian phases or Right-Turn-on-Red (RTOR) volumes were considered for conservative analysis.
- The intersection was analyzed as an isolated intersection without considering downstream blockage or spillback effects.

2.2 New Falls

The second analysis was conducted for the signal design of the New Falls of Neuse Road & Durant Road intersection in North Carolina. The intersection was analyzed for AM, Lunch, and PM peak hour conditions to develop a pre-timed signal plan based on HCM procedures.

Traffic volumes for all movements were provided for each peak period. Lane groups were formed based on turning movements, and capacity analysis was performed using the Critical Lane Method to determine the required cycle length. Signal phasing was designed considering existing geometry, demand, and operational efficiency. Peak Hour Factors (PHF) were assumed due to lack of field data, ranging from 0.85 to 0.95 depending on approach and movement type.

- Base Saturation Flow Rate (S_0) = 1900 passenger cars per hour per lane (pc/h/ln) was used for all lane groups.
- Lane Width Factor (fw) = 1.04 for all approaches.
- Heavy Vehicle Factor (fHVg) varied between 0.9642 and 0.9766 depending on approach.
- Left Turn Factor (fLT) = 0.95 for protected left-turns.
- Right Turn Factor (fRT) = 0.85 for exclusive right-turn lanes.
- Parking Adjustment Factor (fp) = 1.0 (no on-street parking).
- Bus Blockage Factor (fbb) = 1.0 (no transit interference).
- Minimum Green Time = 8 seconds for all phases.
- Yellow Interval = 2 seconds.
- All-Red Interval = 2 seconds.

The Excel file provided on Canvas was used to perform the signal analysis and timing design for the New Falls of Neuse Road and Durant Road intersection. Peak hour volumes for AM, Lunch, and PM were input into the spreadsheet, and built-in formulas were used to calculate flow rates, apply HCM adjustment factors, and identify critical lane groups. The tool also assisted in determining green time distribution for each phase based on traffic demand and lane capacities.

3. Results

3.1 Burlington Mills

The Burlington Mills intersection exhibited efficient operational conditions during the morning (AM) peak period, as summarized in Table 1. The intersection's overall average delay was notably low at 18.8 seconds per vehicle, reflecting LOS B, which indicates favorable performance with minimal congestion. The critical q/s ratio was determined to be 0.382, emphasizing considerable available capacity and efficient use of intersection resources. Individual lane groups generally operated effectively, with most delays below 30 seconds per vehicle.

Table 1. AM flow design for Burlington Mills Intersection

Movement	Lane Group	Volume (q)	Saturation Flow (s)	q/s	Average Delay (sec/veh)	LOS
EBT	Lane Group 1	2.14	1905.3	0.001	54.52	D
WBLT	Lane Group 2	19.65	5587.7	0.004	42.32	D
NBL	Lane Group 3	125.87	1837.9	0.068	27.89	C
SBL	Lane Group 4	21.3	1837.9	0.012	20.27	C
NBTR	Lane Group 5	2094.41	9354.4	0.224	14.28	B
SBTR	Lane Group 6	1759.32	11284.2	0.156	13.17	B
EBR	Lane Group 7	129.41	1614.6	0.08	19.48	B
WBR	Lane Group 8	4.37	1630.2	0.003	15.53	B
EBL	Lane Group 9	312.3	3629.1	0.086	18.24	B

Critical q/s	0.382		Overall Intersection Delay & LOS	18.8 (B)

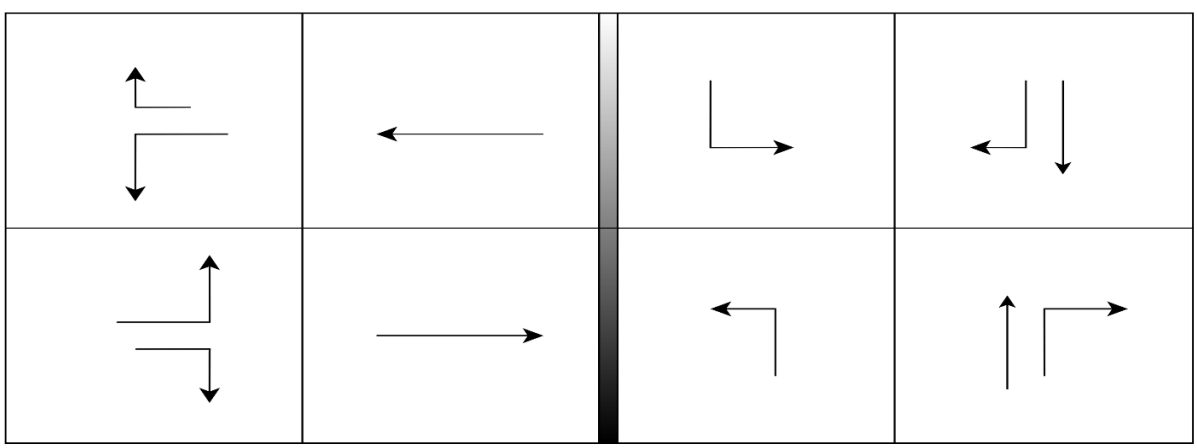

Fig 1. AM phase diagram for Burlington Mills Intersection

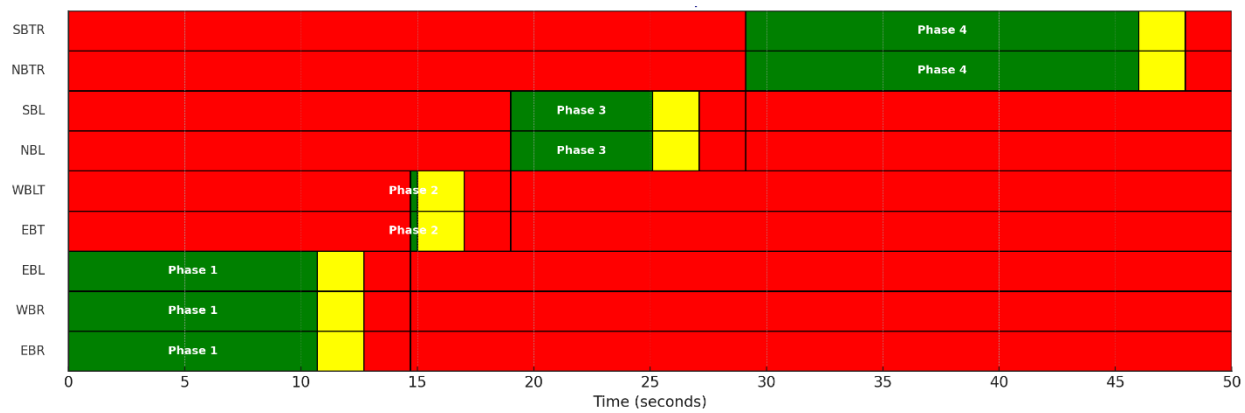

Fig 2. AM phase diagram for Burlington Mills Intersection

Conversely, slightly higher delays were observed in the eastbound through (EBT) at 54.52 seconds per vehicle (LOS D) and the westbound left-turn (WBLT) at 42.32 seconds per vehicle (LOS D), though these delays still remain within acceptable standards for peak-hour traffic. The AM phase diagram provided in Figures 1 and 2 illustrates clear and well-structured movements with minimal conflicting phases, further contributing to the intersection's effective operation.

Table 2. Lunch flow design for Burlington Mills Intersection

Movement	Lane Group	Volume (q)	Saturation Flow (s)	q/s	Average Delay (sec/veh)	LOS
EBL	Lane Group 1	215.64	3629.065143	0.059	19.41	B
EBTR	Lane Group 2	211.17	3519.9	0.060	19.43	B
WBL	Lane Group 3	19.75	3664.068571	0.005	16.69	B
WBTR	Lane Group 4	18.71	3553.8	0.005	16.65	B
NBL	Lane Group 5	145.32	1837.9	0.079	27.11	C
NBTR	Lane Group 6	1439.18	9354.4	0.154	17.49	B
SBL	Lane Group 7	16.20	1837.9	0.009	17.47	B
SBTR	Lane Group 8	1682.87	11284.2	0.149	16.66	B

Critical q/s	0.352		Overall Intersection Delay & LOS	18.37 (B)

The lunchtime analysis of the Burlington Mills intersection revealed highly efficient operational conditions, with an overall intersection delay of 18.37 seconds per vehicle, achieving a favorable LOS B, as detailed in Table 2. The critical volume-to-capacity (q/s) ratio for the intersection was notably low at 0.352, highlighting considerable reserve capacity available for accommodating lunchtime traffic volumes comfortably. Lane group delays were uniformly low, with all movements operating effectively at LOS B or better, except for the northbound left movement, which experienced a slightly higher delay of

27.11 seconds per vehicle but still remained at a satisfactory LOS C. This indicates that even the most constrained movement operates within acceptable standards.

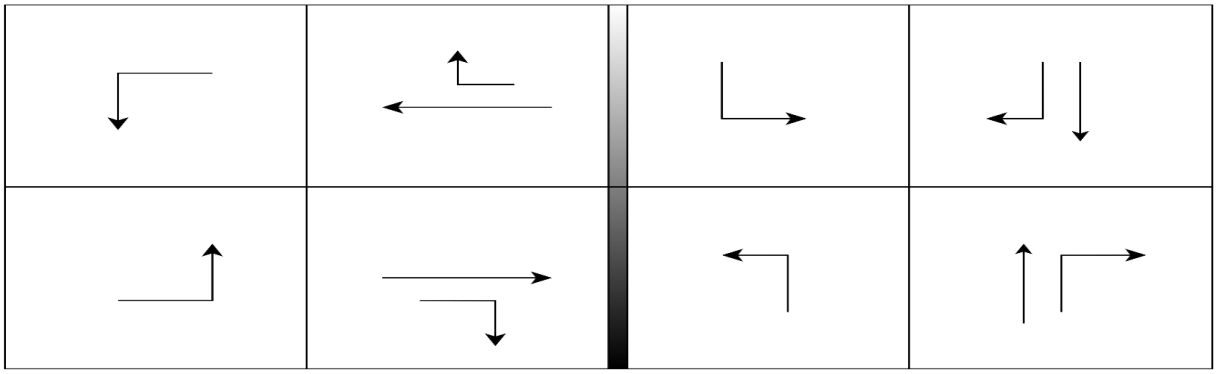

Fig 3. Lunch phase diagram for Burlington Mills Intersection

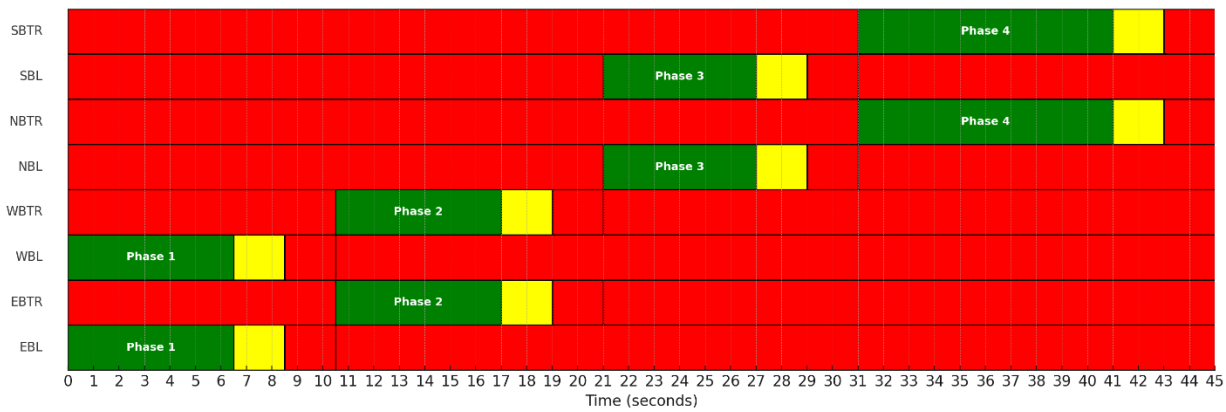

Fig 4. Lunch phase diagram for Burlington Mills Intersection

The lunchtime phase diagram depicted in Figures 3 and 4 confirms a straightforward and efficient arrangement of phases, reducing conflicting movements and promoting smooth vehicle progression through the intersection. Given these findings, the intersection currently demonstrates excellent operational performance during the midday period, requiring minimal intervention beyond standard monitoring.

Table 3. PM flow design for Burlington Mills Intersection

Movement	Lane Group	Volume (q)	Saturation Flow (s)	q/s	Average Delay (sec/veh)	LOS
EBT	Lane Group 1	38.88889	1905.2592	0.020411	89.78	E
WBT	Lane Group 2	35.29412	1923.636	0.018348	42.27	B
NBL	Lane Group 3	490.7749	1837.86819	0.267035	57.46	B
SBL	Lane Group 4	16.07717	1837.86819	0.008748	106.61	E
NBTR-1	Lane Group 5	2497.961	9354.437586	0.267035	49.26	D

NBTR-2	Lane Group 6	1830.452	9354.437586	0.195677	67.92	D
EBR	Lane Group 7	248.6111	1614.626441	0.153974	49.18	D
WBR	Lane Group 8	29.41176	1630.2	0.018042	62.64	D
EBL	Lane Group 9	443.0556	3629.065143	0.122085	52.54	D
WBL	Lane Group 10	35.29412	3664.068571	0.009632	62.28	D
SBTR1	Lane Group 11	1830.452	11284.19919	0.162214	49.96	D
SBTR2	Lane Group 12	2839.966	11284.19919	0.251676	42.03	D

Critical q/s	0.823		Overall Intersection Delay & LOS	54.63(D)

The PM peak operational analysis for Burlington Mills intersection indicated significant capacity constraints, reflected by an overall intersection delay of 54.63 seconds per vehicle, corresponding to LOS D (Table 3). The intersection had a high critical volume-to-capacity (q/s) ratio of 0.823, signifying that the intersection was approaching its operational capacity during the evening period. Notably, certain lane groups encountered substantial delays; specifically, the eastbound through (EBT) and southbound left (SBL) movements experienced the highest delays, at 89.78 seconds (LOS E) and 106.61 seconds (LOS E), respectively. Other movements such as westbound right (WBR), westbound left (WBL), and northbound through/right (NBTR-2) also faced moderate delays ranging between 60 and 70 seconds per vehicle, with LOS D.

The phase structure, as shown in Figures 5 and 6, includes complex, concurrent phases that likely contribute to inefficient traffic flow and higher delays. Given the analysis results, the implementation of optimized signal timing strategies, including protected turn phases and refined cycle splits, is recommended to reduce congestion and improve overall intersection performance during the evening peak period.

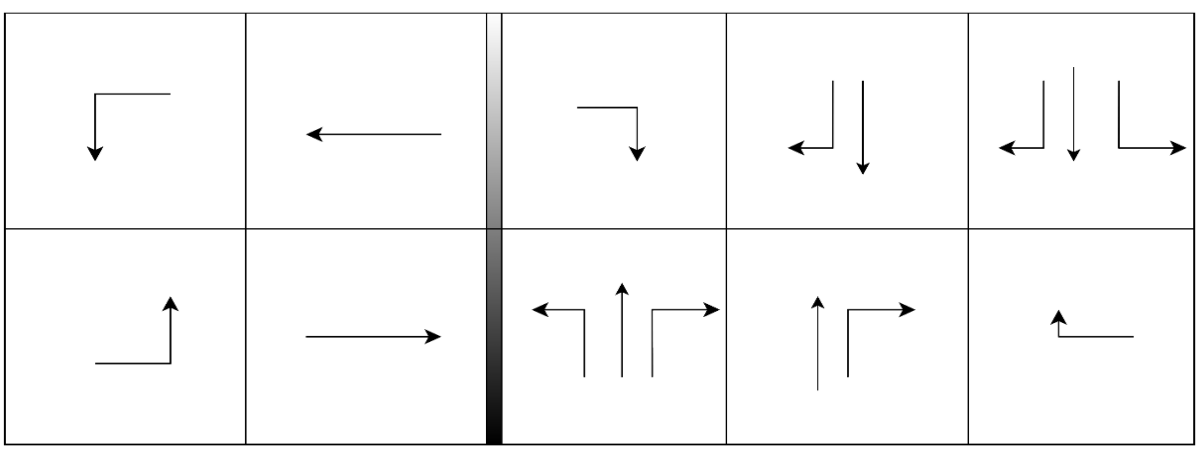

Fig 5. PM phase diagram for Burlington Mills Intersection

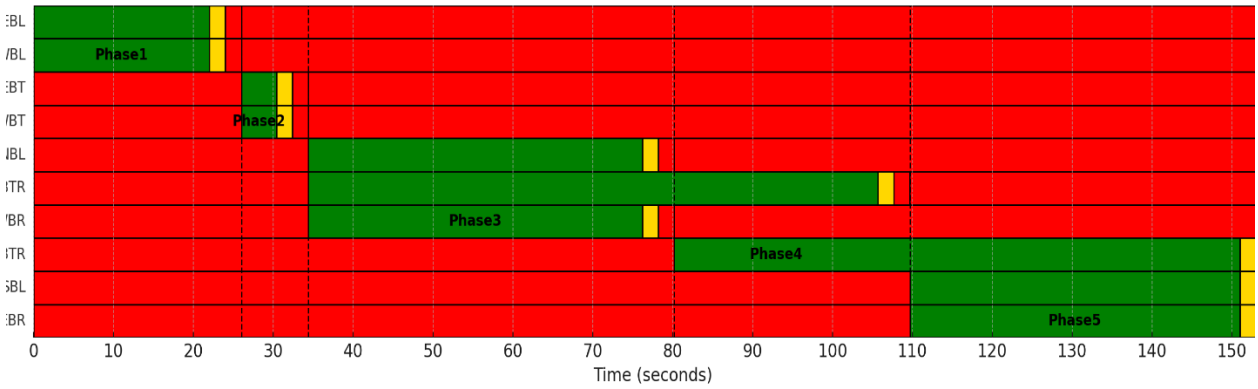

Fig 6. PM phase diagram for Burlington Mills Intersection

3.2 New Falls

The operational analysis for the New Falls of Neuse intersection evaluated traffic conditions during three critical time periods: morning peak, midday, and evening peak. Each period presented distinct operational characteristics, as detailed in Tables 4, 5, and 6, respectively, and visualized through corresponding phase diagrams shown in Figures 4, 5, and 6.

Table 4. AM flow design for New Falls Intersection

Movement	Lane Group	Volume (q)	Saturation Flow (s)	q/s	Average Delay (sec/veh)	LOS
EBT	Lane Group 1	388.24	1900	0.204	56.74	E
EBL	Lane Group 2	300.53	3610	0.083	71.2	E
EBR	Lane Group 3	300.53	1615	0.186		
WBT	Lane Group 4	1174.44	3800	0.309	36.73	C
WBL	Lane Group 5	676.67	3610	0.187	49.39	D
WBR	Lane Group 6	173.33	1615	0.107		
NBT	Lane Group 7	1623.54	5700	0.285	35.94	C
NBL	Lane Group 8	424.24	3610	0.118	56.42	D
NBR	Lane Group 9	347.3	1615	0.215		
SBT	Lane Group 10	927.6	5700	0.163	35.99	C
SBL	Lane Group 11	251.3	3610	0.07	75.49	E
SBR	Lane Group 12	317.4	1615	0.197		

Critical q/s	0.750		Overall Intersection Delay & LOS	63.1 (E)

During the morning peak period, the intersection operated under congested conditions, with an overall average delay of 63.1 seconds per vehicle, classified as LOS E (Table 4). Several lane groups experienced substantial delays, particularly the eastbound left (71.2 sec/veh, LOS E), eastbound through (56.74 sec/veh, LOS E), and southbound left (75.49 sec/veh, LOS E). The critical volume-to-capacity ratio (q/s) of 0.750 indicated significant

saturation, reflecting near-capacity conditions. The phase configuration illustrated in Figures 7 and 8 highlights the complexity of simultaneous left-turn movements, underscoring the need for adjustments such as protected phasing or signal optimization to alleviate these delays.

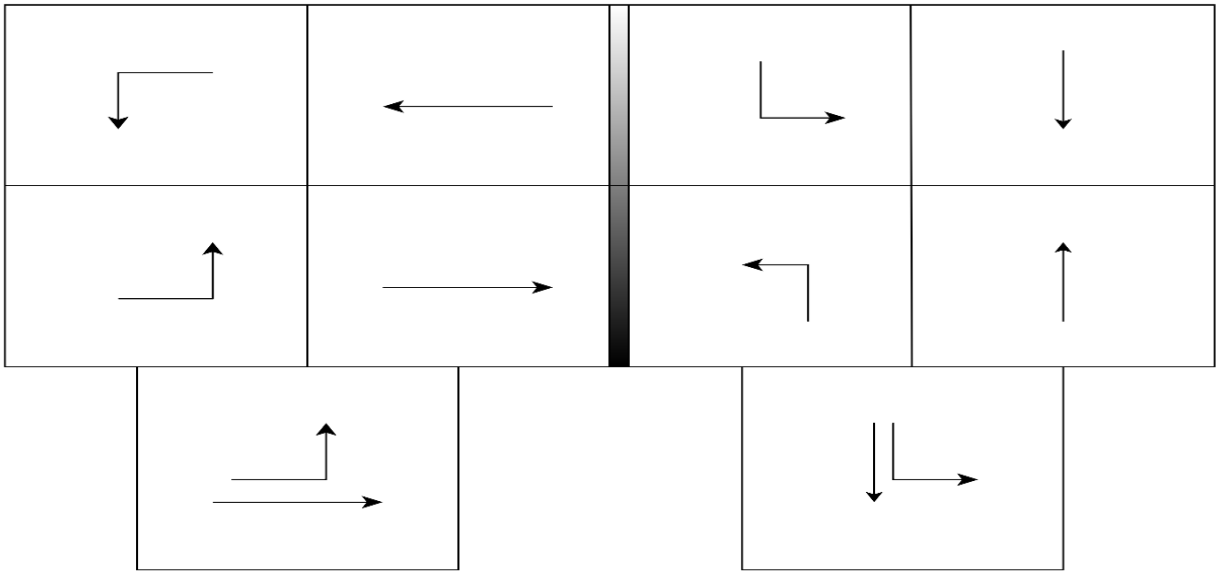

Fig 7. AM phase diagram for New Falls Intersection

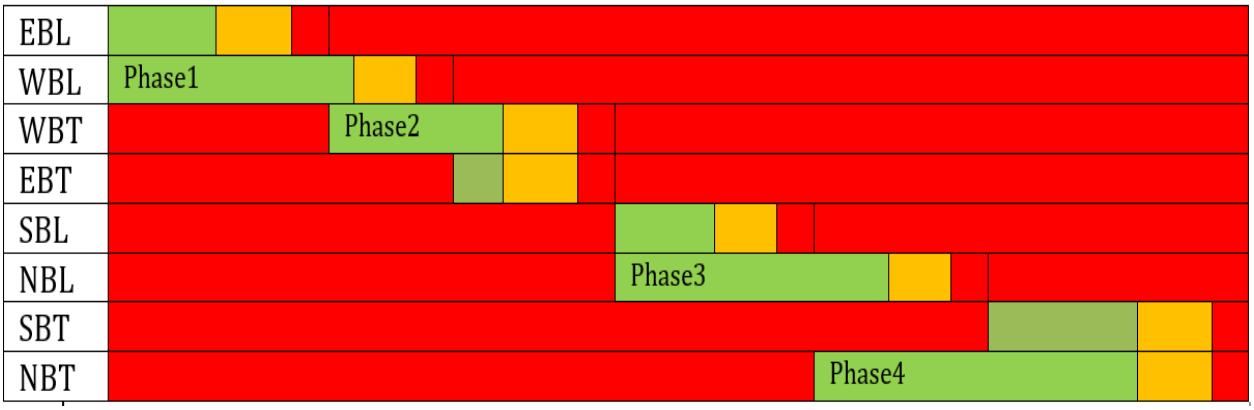

Fig 8. AM phase diagram for New Falls Intersection

In contrast, the midday period showed substantially improved performance, with an average intersection delay of 28.92 seconds per vehicle, achieving LOS C (Table 5). The critical q/s ratio dropped notably to 0.610, indicating adequate capacity to accommodate the traffic volumes comfortably. Most movements operated smoothly, generally maintaining LOS C, except for the northbound left turn (41.87 sec/veh, LOS D) and southbound through movement (54.68 sec/veh, LOS D).

Table 5. Lunch flow design for New Falls Intersection

Movement	Lane Group	Volume (q)	Saturation Flow (s)	q/s	Average Delay (sec/veh)	LOS
EBT	Lane Group 1	393.3	1900	0.207	24.41	C
EBL	Lane Group 2	318.44	3610	0.088	24.36	C
EBR	Lane Group 3	153.07	1615	0.095		
WBT	Lane Group 4	688.89	3800	0.181	31.43	C
WBL	Lane Group 5	557.78	3610	0.155	31.67	C
WBR	Lane Group 6	141.11	1615	0.087		
NBT	Lane Group 7	1196.99	5700	0.21	25.52	C
NBL	Lane Group 8	314.32	3610	0.087	41.87	D
NBR	Lane Group 9	273.4	1615	0.169		
SBT	Lane Group 10	925.9	5700	0.162	54.68	D
SBL	Lane Group 11	142.4	3610	0.039	27.45	C
SBR	Lane Group 12	346.1	1615	0.214		

Critical q/s	0.610		Overall Intersection Delay & LOS	28.92 (C)

Figures 9 and 10 illustrate fewer conflicting phases, reflecting this reduced operational complexity. Thus, midday conditions suggest a favorable operational scenario requiring minimal intervention but continuous monitoring to sustain efficiency.

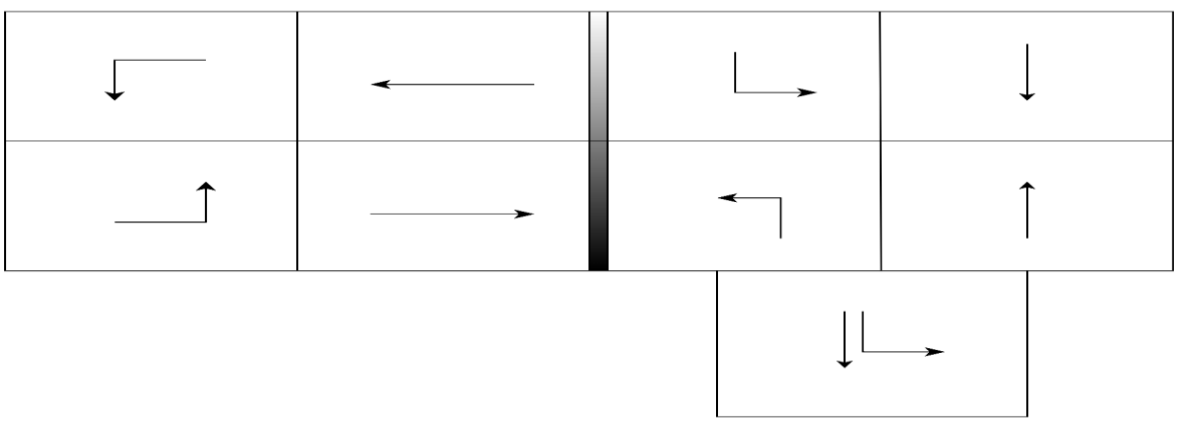

Fig 9. Lunch phase diagram for New Falls Intersection

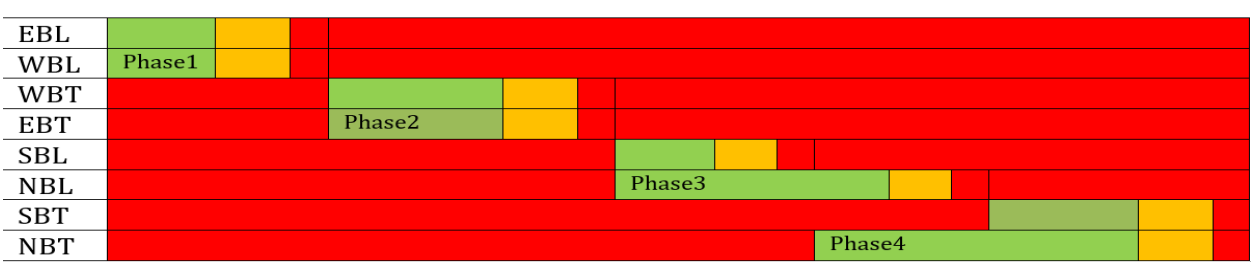

Fig 10. Lunch phase diagram for New Falls Intersection

Table 6. Lunch flow design for New Falls Intersection

Movement	Lane Group	Volume (q)	Saturation Flow (s)	q/s	Average Delay (sec/veh)	LOS
EBT	Lane Group 1	370.53	1900	0.195	48.39	D
EBL	Lane Group 2	371.58	3610	0.1029	77.85	E
EBR	Lane Group 3	127.37	1615	0.0789		
WBT	Lane Group 4	1124.21	3800	0.2958	47.45	D
WBL	Lane Group 5	557.89	3610	0.1545	66.1	E
WBR	Lane Group 6	144.21	1615	0.0893		
NBT	Lane Group 7	1443.16	5700	0.2532	45.17	D
NBL	Lane Group 8	421.05	3610	0.1166	73.84	E
NBR	Lane Group 9	326.32	1615	0.2021		
SBT	Lane Group 10	1526.32	5700	0.2678	47.57	D
SBL	Lane Group 11	221.05	3610	0.0612	53.67	D
SBR	Lane Group 12	437.89	1615	0.2711		

Critical q/s	0.780		Overall Intersection Delay & LOS	63.14 (E)

During the evening peak, the intersection conditions deteriorated again, closely resembling morning peak conditions. The overall average delay was 63.14 seconds per vehicle, indicating LOS E as reported in Table 6. With a critical q/s ratio of 0.780, the intersection operated close to full capacity, highlighting significant congestion issues. High delays were particularly prominent in movements such as eastbound left (77.85 sec/veh, LOS E), westbound left (66.1 sec/veh, LOS E), and northbound left (73.84 sec/veh, LOS E). The phase diagram in Figure 6 illustrates increased complexity in managing high-volume turning movements, suggesting the necessity for enhanced signal strategies like dedicated left-turn phases, optimized cycle lengths, or advanced adaptive control systems during evening peak hours.

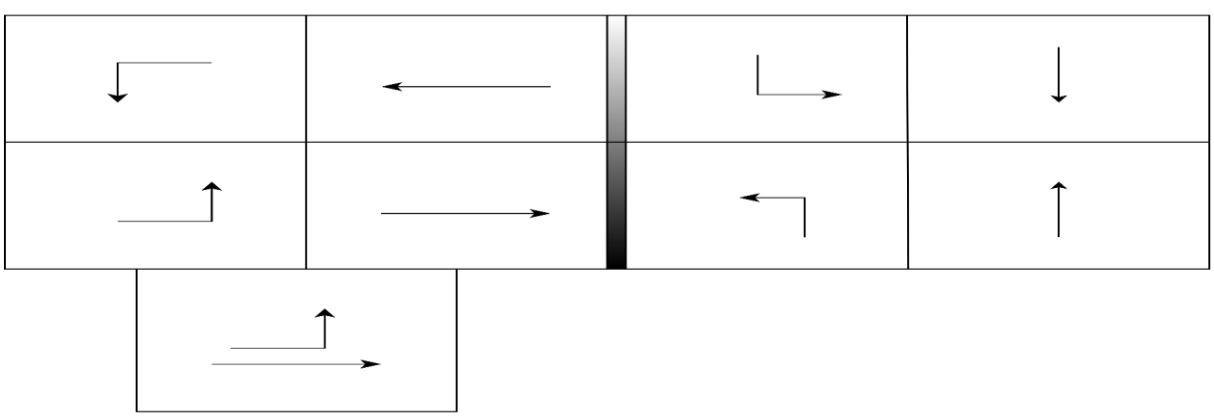

Fig 11. PM phase diagram for New Falls Intersection

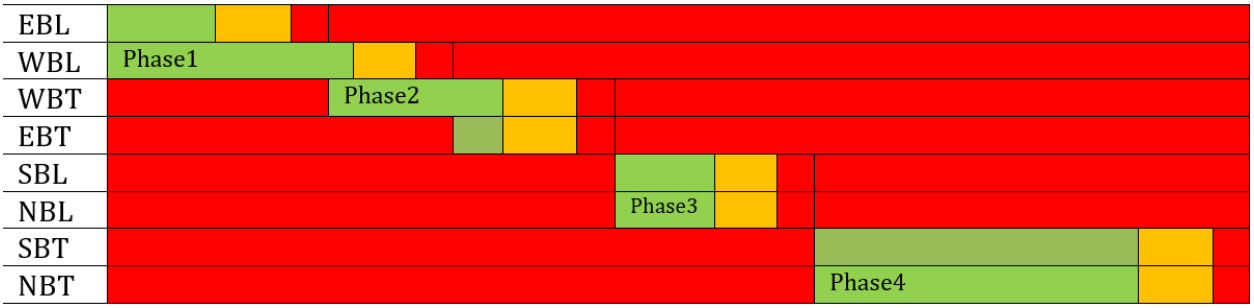

Fig 12. PM phase diagram for New Falls Intersection

4. Conclusion

The pre-timed traffic signal designs developed for Burlington Mills and New Falls intersections using the HCM 7th Edition methodology indicated efficient operational performance during AM and midday peak hours (LOS B–C), characterized by manageable delays and moderate volume-to-saturation (v/s) ratios. However, the PM peak hour analyses revealed significantly increased congestion, with intersection delays rising to 54.63 s/veh (LOS D) at Burlington Mills and 63.14 s/veh (LOS E) at New Falls, alongside critical v/s ratios of 0.823 and 0.780, respectively. These results suggest that, although the proposed designs effectively manage typical traffic conditions, supplementary measures such as actuated or adaptive signal controls could be beneficial for mitigating peak-period congestion and maintaining optimal intersection efficiency.